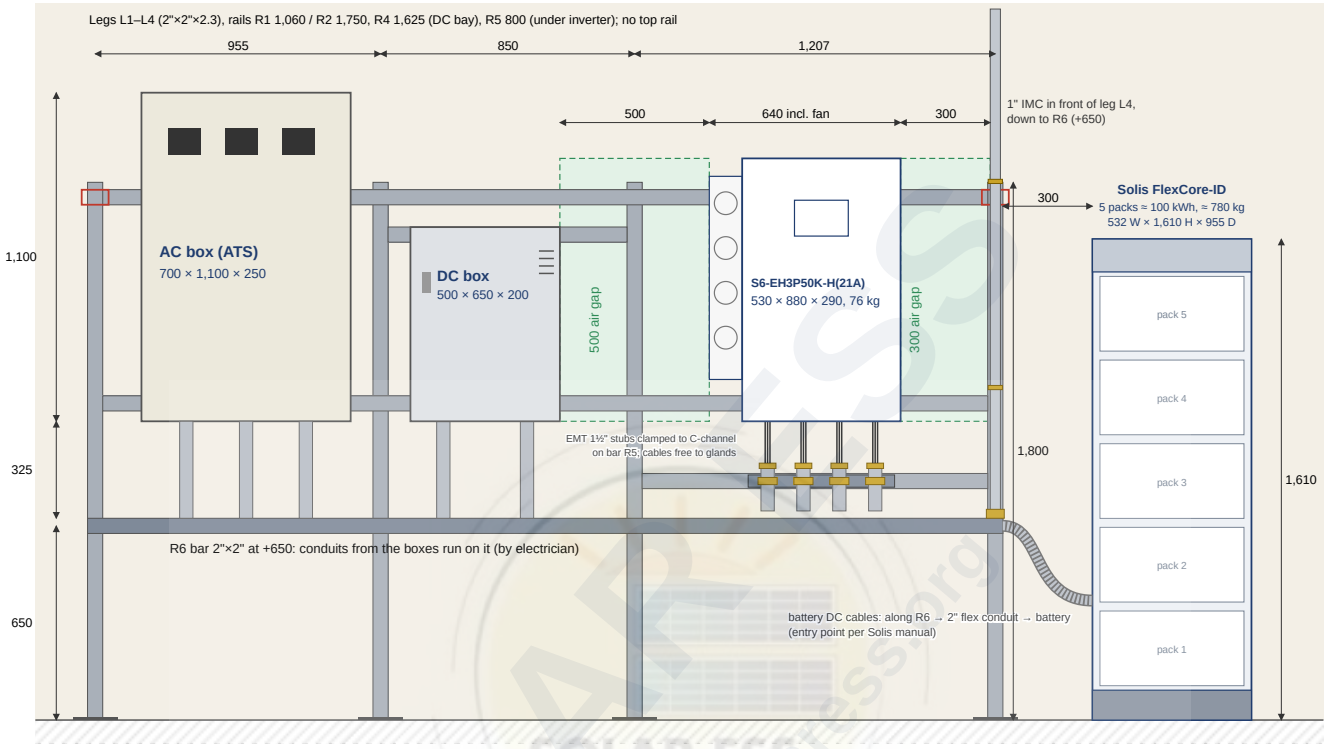
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INSTALLATION MANUAL

50 kW hybrid inverter, AC box (ATS), DC box and 100 kWh battery on a galvanized steel frame

IM-INV-02 · Rev03 · 1 Nov 2026 · materials, cable and breaker schedule, installation plan, tests and handover

Front elevation (from drawing STR-INV-02 Rev21)



System at a glance

Item	Description
Inverter	Solis S6-EH3P50K-H(21A), 3-phase hybrid, 530 × 880 × 290 mm, 76 kg, IP66
Battery	Solis FlexCore-ID, 5 packs ≈ 100.45 kWh, 532 × 1,610 × 955 mm, ≈ 780 kg, on the floor
AC box (ATS)	≈ 700 × 1,100 × 250 mm: Q1 main (inverter backup), Q2 generator, Q3 GEN port, 4-pole ATS
DC box	≈ 500 × 650 × 200 mm: PV string protection and isolation
Frame	2"×2"×2.3 galvanized steel, 4 legs, 500 mm off the wall on 2 wall arms; bar R6 at 650 carries the conduits

Contents

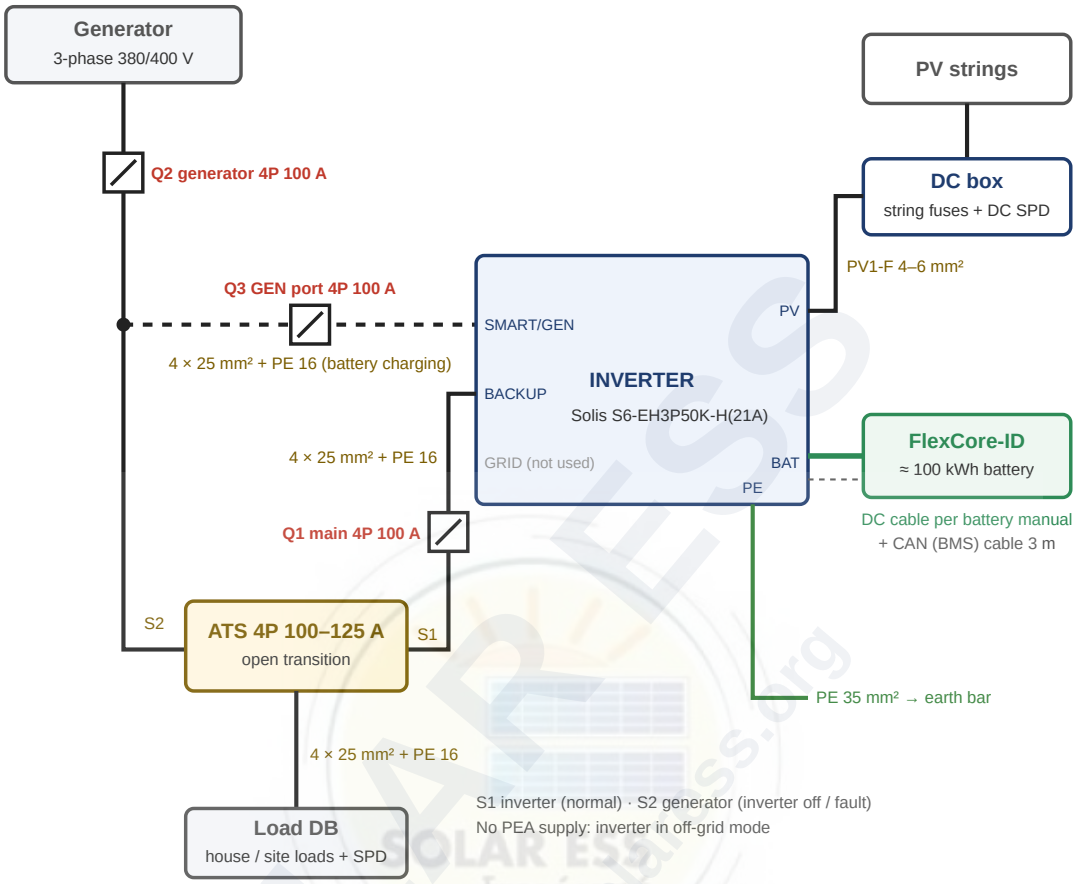
1. Cover and system overview
2. Single-line diagram, cable and breaker schedule
3. Materials list A: structure and containment
4. Materials list B: electrical, earthing, battery, tools
5. Installation plan: sequence, crew and hold points
6. Step-by-step installation
7. Tests, commissioning and handover
8. Appendix: drawing STR-INV-02 Rev21 (2 sheets)

Key rules (Solis manual): ≥ 500 mm clear left, right and top; ≥ 1,000 mm inverter bottom to floor and in front; vertical ± 5°; structure ≥ 4 × weight; no PEA supply, so the Grid port is not used; the generator never runs in parallel with the Backup output.

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1. Single-line diagram and cable schedule

A. Single-line diagram (off-grid site, no PEA supply: inverter backup and generator through the ATS)



B. Cable and breaker schedule (copper, in conduit, 380/400 V)

Circuit	From → to	Cable	Breaker / device	Solis (21A) manual
Backup (main)	Inverter BACKUP → Q1 → ATS S1	4 × 25 mm² + PE 16 mm²	Q1: 4P MCCB 100 A, ≥ 25 kA	Backup 20–25 mm²; breaker ≥ 125 A (see note)
Generator	Generator → Q2 → ATS S2	4 × 25 mm² + PE 16 mm²	Q2: 4P MCCB 100 A (to suit generator)	—
GEN port	Q2 load side → Q3 → SMART/GEN	4 × 25 mm² + PE 16 mm²	Q3: 4P MCCB 100 A	Smart port 20–25 mm²
Load feed	ATS → load DB	4 × 25 mm² + PE 16 mm²	ATS 4P 100–125 A, open transition	Backup never paralleled
Grid port	Not used (no PEA supply)	—	—	Grid port 50 mm² if added later
Earth	Inverter PE → earth bar	35 mm² green/yellow	—	33 mm²
PV	DC box → inverter PV	PV1-F 4–6 mm²	String fuses, DC SPD	4–6 mm², strip 7 ± 0.5 mm
Battery	FlexCore-ID → inverter BAT	Per battery manual	Battery DC breaker per manual	Battery cable 10–25 mm²

Breaker note: the (21A) manual asks for a Backup breaker ≥ 125 A, but its terminal takes only 20–25 mm² (≈ 90–100 A in conduit). We fit 100 A on 25 mm², which covers 76 A rated and rides through the 1.6 × short peaks. Confirm with Solis Thailand before ordering; a 125 A breaker needs a larger cable with a Solis-approved reducing lug.

Rules: ATS breaks before it makes and switches all 4 poles (inverter and generator each keep their own N–PE bond). RCD loads on the backup side ≤ 60 % of the inverter rating. Phase colours: L1 brown, L2 black, L3 grey, N blue, PE green/yellow. Check voltage drop for runs over 30 m. Cable gland torque 7–7.5 N·m; terminal torques per the inverter label.

2. Materials list

Materials list A: structure, fixings and containment

Ref	Item	Specification	Qty	Unit	Notes
1. Steel frame (galvanized after welding)					
L1–L4	Legs, top capped	2"×2"×2.3 SHS, 1,800 mm	4	pcs	Spacing 955 / 850 / 1,207
R1–R2	Rails at 1,060 and 1,750	2"×2"×2.3 SHS, ≈ 3.06 m	2	pcs	Through-bolt holes to suit boxes
R4	DC box rail at 1,625	2"×2"×2.3 SHS, ≈ 0.80 m	1	pc	Level with DC box top
R5	EMT bar under inverter at 800	2"×2"×2.3 SHS, ≈ 1.16 m	1	pc	Carries C-channel C1
M1–M2	Verticals for Solis bracket	2"×2"×2.3 SHS, ≈ 0.64 m	2	pcs	≈ 440 apart, centred on inverter
R6	Conduit bar at 650 (front of legs)	2"×2"×2.3 SHS, ≈ 3.06 m	1	pc	Replaces the wireways
A1–A2	Wall arms behind L1 and L4 at 1,750	2"×2"×2.3 SHS, 492 + plate	2	pcs	Legs 500 mm off the wall
—	Steel stock to order	2"×2"×2.3 SHS, 6 m lengths	4	lengths	≈ 20.7 m used, ≈ 72 kg
P4	Base plates	150 × 150 × 8 SS400	4	pcs	Welded to legs
P5	Wall plates on the arms	100 × 100 × 8, 2 × Ø12	2	pcs	2 × M10 each
2. Fixings					
—	Floor anchors	M10 × 80 wedge anchor, HDG/SS	8	pcs	2 per base plate
—	Wall anchors	M10 × 80 wedge anchor, HDG/SS	4	pcs	2 per wall plate
—	Inverter bracket bolts	M8 × 80 through-bolt, nyloc nut, 2 large washers	5	sets	3 in R2, 2 in M1/M2
—	Box fixing bolts	M8 × 70 through-bolt, nyloc, washers	12	sets	AC box 4, DC box 4, spare 4
—	Inverter bracket	Solis wall bracket 497 × 452.5 + locking screws	1	set	Supplied with inverter
3. Containment					
C1	C-channel (strut)	41 × 41 × 2.5 HDG	0.5	m	Bolted to R5, 2 × M10
—	Strut conduit clamps	For 1½" EMT	4	pcs	On C1
E1	EMT stubs to inverter	1½" EMT, 150 mm, connector + locknut	4	pcs	Cables free above to glands
E3	EMT into AC box (3) and DC box (2)	1½" EMT, 300 mm, connectors + locknuts	5	pcs	All the way into the boxes
E2	IMC drop at L4	1" IMC, connector + locknut, straps	1	run	Down to R6
E4	Battery flexible conduit	2" flex, 2 connectors	≈ 1	m	Along R6 → battery
—	Conduits on R6 and to the generator / load DB, with straps	Per route	per	site	Measure on site

2. Materials list (continued)

Materials list B: electrical, earthing, battery, consumables and tools

Ref	Item	Specification	Qty	Unit	Notes
4. AC protection and switching (in AC box)					
Q1	Main breaker (inverter Backup output)	MCCB 4P 100 A, ≥ 25 kA	1	pc	See breaker note p. 2
Q2	Generator breaker	MCCB 4P 100 A, ≥ 25 kA	1	pc	Size to the generator
Q3	GEN port breaker (recommended)	MCCB 4P 100 A, ≥ 25 kA	1	pc	Generator charges the battery
ATS	Automatic transfer switch	4P 100–125 A, open transition, with controller	1	set	S1 inverter, S2 generator
SPD	AC surge protector (recommended)	Type 2, 4P, 40 kA, 385 V	1	set	Load side of ATS, with its own breaker
—	Busbar / terminal blocks, DIN rail, labels	To suit AC box	1	lot	
5. Cables (copper, phase colours brown / black / grey / blue / green-yellow)					
—	Backup / generator / GEN port / load cables	4 × 25 mm² per circuit (THW single cores or CV)	per	site	≈ 2 m inside the AC box per circuit
—	Protective earth	16 mm² (25 mm² circuits), 35 mm² (inverter PE)	per	site	Green/yellow
—	Bonding	16 mm² green/yellow	≈ 10	m	Frame, conduits, battery cabinet
—	PV cable	PV1-F 4–6 mm², MC4	per	string	DC box → inverter
—	Cable lugs	Copper tube lugs 50 / 25 / 16 / 35 mm², sizes to terminal bolts	1	lot	Inverter supplies 13 OT terminals
6. Battery, communication and earthing					
B1	Battery system	Solis FlexCore-ID, 5 packs ≈ 100 kWh	1	set	Floor, 300 right of L4
—	Battery DC cables and breaker	Per Solis FlexCore-ID manual	1	set	In 2" flex E4
—	CAN / BMS cable	Supplied, 3 m	1	pc	Battery → inverter BMS 1
—	Anti-tip anchors for battery	Per battery manual	1	set	
—	Earth bar and earth electrode	Copper bar; rod 5/8" × 2.4 m if none exists	1	set	Earth ≤ 5 Ω
7. Consumables					
—	Cold galvanizing paint	Zinc-rich spray	2	cans	On all welds and cut ends
—	Heat shrink, cable ties (UV), markers, warning labels	Assorted	1	lot	Label every breaker and cable
—	Fire stopping	Intumescent sealant	1	tube	Wall / floor penetrations

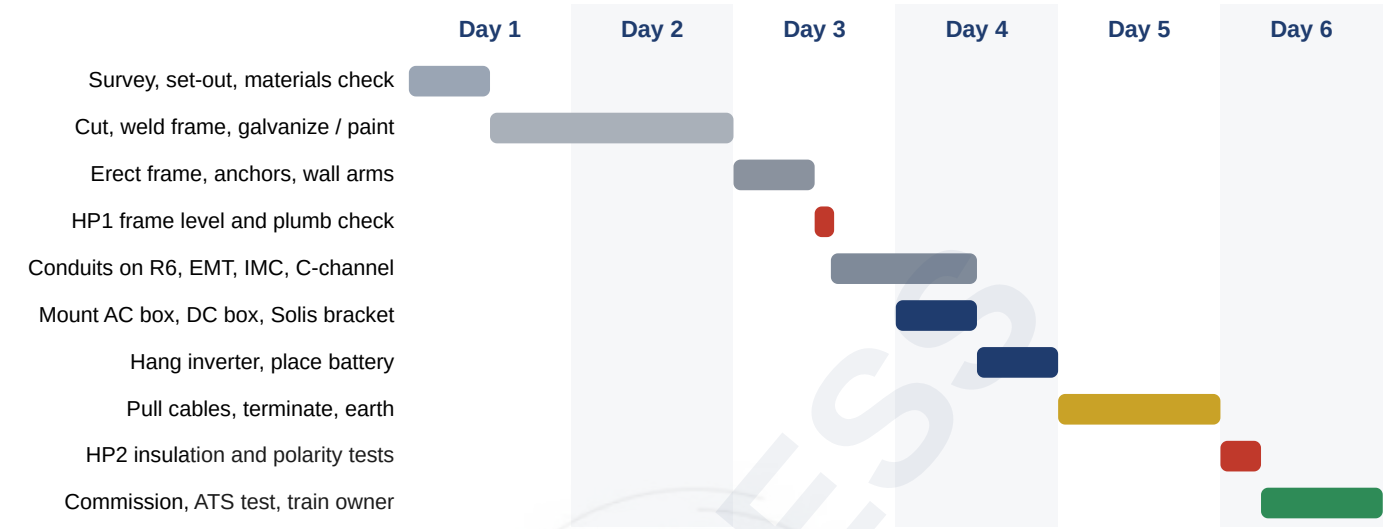
8. Tools and test instruments

Welder and grinder · drill, Ø9 / Ø11 bits, hammer drill for anchors · spirit level and laser · torque wrench (to 30 N·m) · hydraulic crimper to 50 mm² · cable cutter · insulation tester 1,000 V · multimeter and clamp meter · phase rotation meter · earth resistance tester · lifting straps (two-person lift, 76 kg inverter) · pallet jack or trolley for the 780 kg battery · PPE and lock-out / tag-out kit

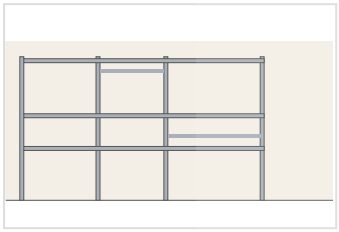
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3. Installation plan

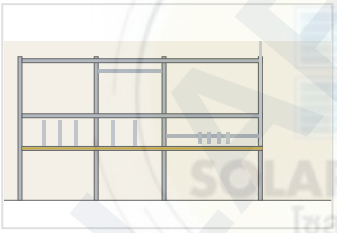
A. Installation sequence (about 6 working days)



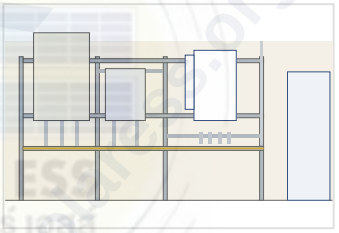
B. Build stages



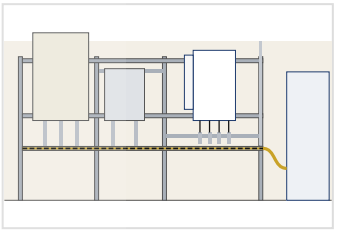
Stage 1
Frame erected and tied



Stage 2
Conduits on R6, EMT, IMC



Stage 3
Boxes, inverter, battery



Stage 4
Cables pulled and terminated

C. Crew

Role	No.	Main tasks
Site lead / electrician (licensed)	1	Set-out, terminations, tests, commissioning
Electrician	1	Containment, cable pulling, labelling
Welder / fitter	1	Frame, brackets, anchors (days 1–3)
Helper	1	Lifting (inverter 2 people), material handling

D. Hold points

Hold	Check before continuing
HP1	Frame plumb and level ± 2 mm, all anchors and ties tight, galvanizing repaired
HP2	Insulation ≥ 1 MΩ at 1,000 V, PV polarity, phase rotation, earth ≤ 5 Ω, all torques marked
HP3	ATS transfer both ways between inverter and generator, never paralleled

Before ordering: confirm the AC and DC box sizes, the Backup breaker rating with Solis Thailand, the cable routes and lengths, and widen the inverter's right-side gap from 300 to the Solis 500 mm (moves leg L4, the IMC and the battery 200 mm right).

4. Installation steps

Step-by-step installation

1 Set out
Mark leg centres 955 / 850 / 1,207 on the floor and wall. Keep the back of the legs 500 mm off the wall (air gap). Check the wall can take the 2 arm plates and the floor slab the 780 kg battery (≈ 15 kN/m²).

2 Fabricate
Cut legs, rails R1, R2, R4, R5 and verticals M1/M2; weld base plates; drill box and bracket holes Ø9 before galvanizing or paint welds with cold galvanizing.

3 Erect frame
Stand legs plumb, fix base plates (2 × M10 each), weld the 2 wall arms behind L1 and L4 at 1,750 and anchor their plates (2 × M10 each). Hold point HP1.

4 Containment
Strap the conduits to R6 (+650); C-channel C1 on R5; IMC 1" down in front of L4 to R6.

5 AC and DC boxes
Bolt the AC box (bottom at 1,000) and the DC box (top on R4 at 1,625). Fit EMT 1½" from R6 up into the boxes (3 + 2).

6 Inverter bracket
Bolt the Solis bracket: 3 top holes through R2, 2 bottom holes through M1/M2 (5 × M8 × 80). Check level ± 5°; inverter bottom will sit at 1,000.

7 Hang inverter
Two people lift the 76 kg inverter by its handles onto the bracket; fit the Solis locking screws. Keep 500 mm clear on the fan side and on top.

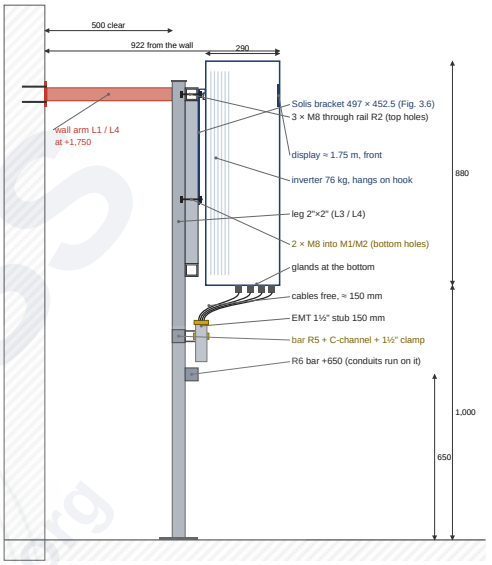
8 Battery
Move the FlexCore-ID (≈ 780 kg) on a pallet jack to 300 right of L4; level and fix the anti-tip anchors; fit the 2" flex conduit from the end of R6.

9 Cables
Pull backup, generator, GEN port and load cables 4 × 25 and PE cables; PV and battery DC cables in their own runs. EMT stubs to inverter clamped on C1, cables free to the glands.

10 Terminate
Crimp lugs, tighten to the torques on the inverter label; glands 7–7.5 N·m. Label every cable both ends and every breaker.

11 Earth
Inverter PE 35 mm² to the earth bar; bond frame, conduits and battery cabinet with 16 mm². Off-grid site: drive an earth electrode if none exists; measure earth ≤ 5 Ω.

12 Test and commission
Hold point HP2 tests, then power up per page 7; ATS test (HP3); set up the app and monitoring; hand over.



Side section through the inverter (STR-INV-02 sheet 2): bracket on R2 and M1/M2, EMT on C1, R6 at 650, legs 500 mm off the wall.

Heights
R6 650 · equipment bottoms 1,000 · inverter top 1,880 · display ≈ 1.75 m · legs 1,800 · battery 1,610

5. Tests, commissioning and handover

A. Tests before energising (HP2)

	Test	Pass value
☐	Visual: torques marked, glands tight, covers on	No loose parts
☐	Insulation resistance, AC circuits, 1,000 V	≥ 1 MΩ
☐	Insulation resistance, PV strings	≥ 1 MΩ
☐	PV string open-circuit voltage and polarity	Within design, + / − correct
☐	Earth electrode and PE continuity	≤ 5 Ω; continuity < 0.5 Ω
☐	Phase rotation at inverter output and generator (same order)	L1-L2-L3 clockwise
☐	Battery voltage and BMS link	Per battery manual

B. Power-up sequence (Solis)

1. Battery breaker / switch ON; check the BMS.
2. DC switch on the inverter ON; set off-grid mode in the Solis app.
3. Q1 main ON; ATS to S1 (inverter).
4. Q2 and Q3 ON; start the generator and check it charges the battery through the GEN port.
5. Set battery type, off-grid work mode, generator start / stop SOC in the Solis app.
6. The backup side stays off until commissioning is complete.

C. Commissioning checks (HP3)

	Check	Result
☐	Inverter in off-grid mode, no alarms	
☐	PV power reading on every MPPT	
☐	Battery charges and discharges	
☐	Generator charges battery via GEN port	
☐	Inverter off: ATS moves loads to generator and back	never paralleled
☐	Backup load ≤ rated; RCD loads ≤ 60 %	
☐	Monitoring online, owner app set up	

D. Handover

	Item	
☐	As-built drawings and this manual	
☐	Test results signed	
☐	Breaker and cable labels, warning signs	
☐	Owner trained: start-up, shut-down, ATS bypass	
☐	Warranty registration (inverter and battery)	

E. Safety

> Isolate and lock out AC, PV and battery before work; PV strings are live in daylight. Two-person lift for the inverter; pallet jack for the battery. Never run the generator in parallel with the Backup output; set one N–PE bond per source as Solis and the generator maker require. Keep 500 mm clear around the inverter and nothing on top of it. Work by licensed electricians to EIT (วสท.) standards.

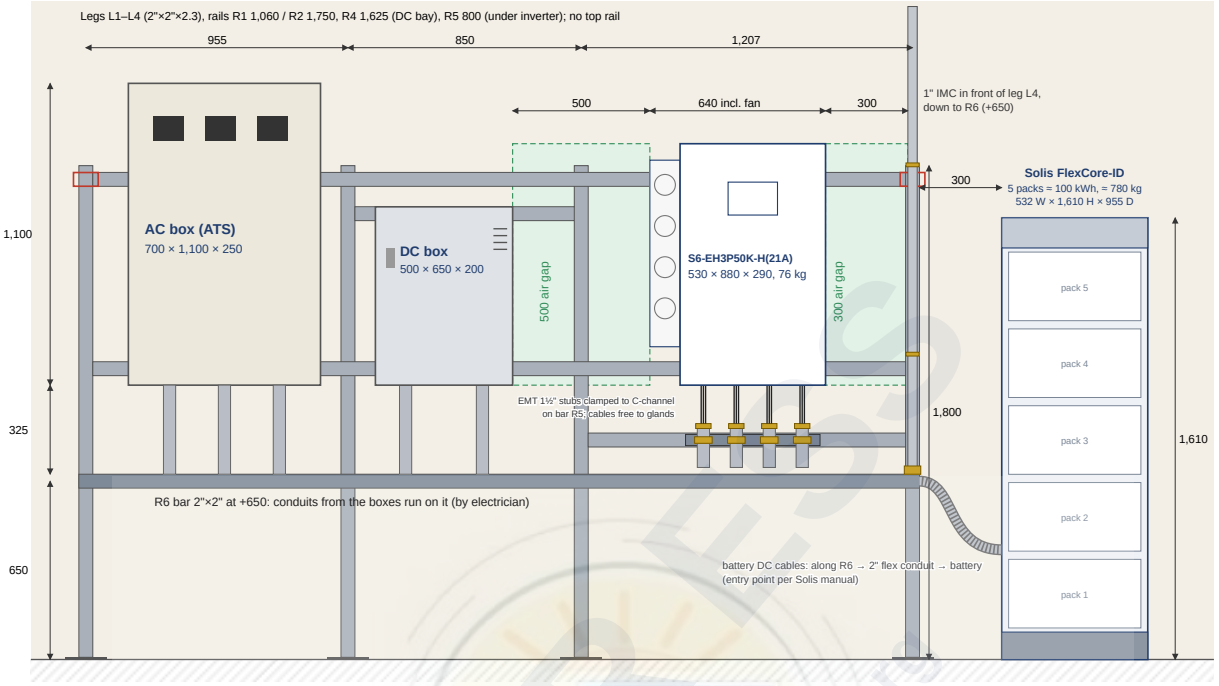
Solis user manual S6-EH3P(50-60)K-H(21A), V1.2 (16 Jul 2026): https://www.solisinverters.com/uploads/file/Solis_Manual_S6-EH3P(50-60)K-H(21A)_EUR_V1,2(20260716).pdf. Appendix: drawing STR-INV-02 Rev21 (2 sheets) follows this page.

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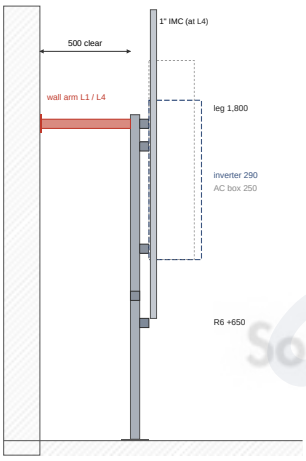
STR-INV-02 · EQUIPMENT FRAME + FLOOR BATTERY

AC box, DC box and Solis S6-EH3P50K-H(21A) on one 2"×2"×2.3 frame 500 mm off the wall; FlexCore-ID battery on the floor · mm

A. Front elevation



B. Side section



C. Parts (whole frame)

Ref	Part	Size	Qty	Remarks
L1–L4	Steel legs, 1,800 mm, top capped	2"×2"×2.3	4	955 / 850 / 1,207
M1–M2	Verticals R1–R2 for the Solis bracket bottom holes	2"×2"×2.3	2	≈ 0.64 m, 440 apart
R1–R2	Rails at 1,060 and 1,750, from L1 to L4 (≈ 3.06 m); no top rail	2"×2"×2.3	2	Set to box holes
R4	Short rail at 1,625, level with the DC box top	2"×2"×2.3	1	DC bay (L2–L3)
R5	Bar L3–L4 at 800, under the inverter	2"×2"×2.3	1	≈ 1.16 m
C1	C-channel on R5 + 4 × 1½" EMT clamps	41×41×2.5	0.5 m	Holds the EMT stubs
R6	Bar at 650 on the front of the legs, carries the conduits (replaces the wireways)	2"×2"×2.3	1	≈ 3.06 m
A1, A2	Wall arms behind L1 and L4 at 1,750, 492 + wall plate 100×100×8 (2 × M10): legs 500 mm off the wall	2"×2"×2.3	2	500 mm air gap
P4	Base plate 150×150×8 + 2 × M10 floor anchors	SS400	4	Welded to legs
E2	1" IMC drop in front of leg L4 down to R6, strapped to R1 and R2	IMC 1"	1	Route on site
E3	EMT 1½" from R6 up into the AC box (3) and DC box (2), ≈ 325 mm	EMT 1½"	5	All the way in
E1	EMT 1½" stubs 150 mm, half-way up to the inverter	EMT 1½"	4	Cables free to glands
B1	FlexCore-ID battery, 5 packs ≈ 100.45 kWh, on the floor 300 right of L4; anti-tip anchors per manual	532×1,610×955	1	≈ 780 kg, ≈ 15 kN/m² on the slab
E4	Battery DC cables along R6 into the battery	Flex 2"	≈ 1 m	Cable size per Solis manual
—	Steel ≈ 20.7 m 2"×2" (4 × 6 m) ≈ 72 kg + plates; bolts M8/M10, anchors 8 floor + 4 wall (welder job sheet WJ-STR-INV-02)		set	HDG after welding

Notes

1. Legs stand 500 mm off the wall (air gap) on 2 wall arms. R6 at 650 carries the conduits; all units bottom at 1,000. Inverter display ≈ 1.75 m.
2. Left to right: AC box, DC box (centre), inverter, battery. Inverter gaps: 500 fan side (L3 in it), 300 to the 1" IMC at L4.
3. Box sizes from photos (AC 700 × 1,100 × 250, DC 500 × 650 × 200); set legs/rails to the actual boxes. Inverter 530 × 880 × 290, 76 kg (Solis manual V1.2): https://www.solisinverters.com/uploads/file/Solis_Manual_S6-EH3P(50-60)K-H(21A)_EUR_V1,2(20260716).pdf
4. All EMT 1½" (9 runs): inverter stubs 150 mm clamped to C1 on R5, cables free to glands; AC and DC boxes EMT all the way in.
5. Earth frame, conduits and battery with 16 mm²; cap tube ends.

Strength check (AC box ≈ 80 kg, inverter 76 kg, DC box ≈ 25 kg, conduits and cables ≈ 30 kg; × 1.5)

1. Rails over 1,207 mm (inverter bay) and 955 mm (AC bay): M ≈ 0.2 kN·m → ≈ 30 MPa in 2"×2". **OK, ≈ 5×**
2. Legs: ≈ 0.8 kN each → < 2 MPa. **OK**
3. Equipment hangs in front of the frame: ≈ 0.8 kN·m, taken by 2 wall arms at 1,750 (L1, L4) and the floor anchors, ≈ 0.2 kN per arm vs ≈ 5 kN per M10. **OK**

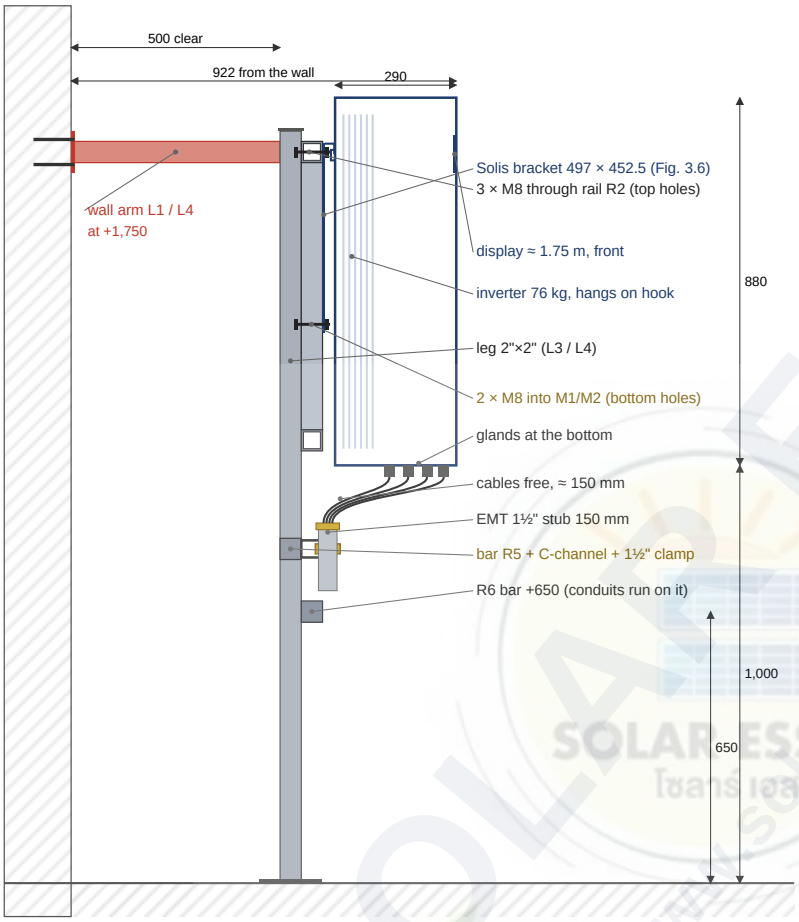
Battery (B1): stands on its own base on a level concrete floor, ≈ 780 kg on 0.51 m² ≈ 15 kN/m², fine on a ground slab (check if the floor is suspended). Keep clearances, front access and cable entry per the Solis manual.

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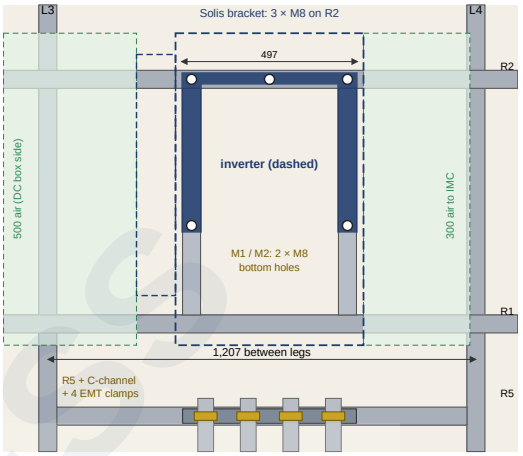
STR-INV-02 · INVERTER SIDE INSTALLATION

How the Solis S6-EH3P50K-H(21A) (530 × 880 × 290 mm, 76 kg, per Solis manual V1.2) is hung on the frame · mm

D. Side section through the inverter



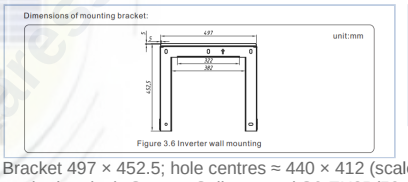
E. Front of the inverter bay, L3–L4



Load check (76 kg × 1.5)

1. Shear on 5 × M8 (3 in R2, 2 in M1/M2): ≈ 0.22 kN per bolt vs ≈ 7 kN. **OK**
2. Inverter centre ≈ 180 mm in front of the rails: ≈ 0.2 kN·m, taken by the bracket holes (≈ 412 apart) ≈ 0.5 kN pull. **OK**
3. Rails R1/R2 over 1,207 mm: ≈ 20 MPa. **OK**

F. Solis bracket and clearances





Bracket 497 × 452.5; hole centres ≈ 440 × 412 (scaled from Fig. 3.6, confirm on the bracket). Source: Solis manual S6-EH3P(50-60)K-H(21A) V1.2 (link below).

Clearance	Solis	This frame	
Fan side (left, DC box)	≥ 500	500	**OK**
Right side (IMC)	≥ 500	300	**Short 200**
Top	≥ 500	open	**Keep clear**
Bottom to floor	≥ 1,000	1,000	**OK**
Front	≥ 1,000	site	Keep walkway clear

Also per Solis: vertical ± 5°, nothing on or against the inverter, structure good for 4 × its weight (≈ 300 kg; this frame OK).

G. Installation steps

1. Frame first (welder job sheet WJ-STR-INV-02): legs plumb 500 mm off the wall, base plates anchored, 2 wall arms at 1,750 tight, rails level (± 2 mm).
2. Weld verticals M1/M2 (2"×2") between R1 and R2 at the bracket's leg holes (≈ 440 apart, centred on the inverter), and bar R5 at 800 with C-channel C1.
3. Hold the Solis bracket (Fig. 3.6) with its 3 top holes on R2 and 2 bottom holes on M1/M2; inverter 300 clear of the IMC, 500 of the DC box. Drill Ø9 through the tubes.
4. Fit 5 × M8 × 80 through-bolts, large washers, nyloc nuts; check level. Set so the inverter bottom is at 1,000.
5. Two people lift the inverter (76 kg) by its handles and hook it onto the bracket; never lift by the glands or fan.
6. Fit the Solis locking screws between the inverter and the bracket.
7. Fix the 1½" EMT stubs (150 mm) with strut clamps on C1, run the cables from the conduits on R6 through them, then free into the bottom glands; keep bend radius and drip loops.
8. Earth the inverter and frame (16 mm²); keep the 500 mm (fan side) and 300 mm (IMC side) air gaps clear.

Solis user manual S6-EH3P(50-60)K-H(21A), V1.2 (16 Jul 2026): https://www.solisinverters.com/uploads/file/Solis_Manual_S6-EH3P(50-60)K-H(21A)_EUR_V1.2(20260716).pdf

Check bracket holes (Fig. 3.6) and torques against it before drilling. Right-side gap 300 is below the Solis 500: widen before construction.